

LESSON 4 Least Squares via Householder QR Factorization

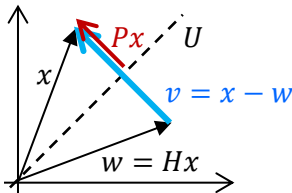
The least squares solution of $Ax = b$ may be obtained by solving the normal equation $(A^T A)x = A^T b$. However, when we use computers for matrix calculations, we need to consider the **condition number** of a matrix which is a number between 1 and ∞ that measures the stability of the matrix. It is known that $\text{cond}(A^T A) = (\text{cond } A)^2 \geq \text{cond } A$, and $A^T A$ can be much more unstable than A . This is a topic in numerical linear algebra, we will not go further, interested students may consult Lesson 8 of the following notes:

<https://lplee.sites.northeastern.edu/math-3530-numerical-analysis/>

In this lesson, we will present a method to find the least squares solution that avoids $A^T A$ using **QR factorization** of A . We first decompose $A = QR$, where Q is orthonormal and R is upper triangular, using an efficient and numerically stable algorithm by **Householder**, (not Gram-Schmidt in MATH2331 that is conceptually simpler but less stable numerically).

4.1 Householder Reflectors

In least squares problems coming from data fitting, we have an $m \times n$ matrix A with n independent columns and $m > n$ (so that $Ax = b$ is inconsistent and over-determined). Householder QR factorization involves the computation of a series of **Householder reflectors**. In this context, a **Householder reflector** H is an $m \times m$ matrix representing a linear map $T: \mathbb{R}^m \rightarrow \mathbb{R}^m$ that reflects vectors in $\mathbb{R}^m$ over an $m - 1$ dimensional subspace U (a hyperplane). Specifically, given two vectors x and w of the same length, we want to find H such that $Hx = w$. Let $v = x - w$, then $v = 2Px$, where Px is the **projection** of x onto the span of v .



So $Hx = w = x - v = x - 2Px = (I - 2P)x$, and $H = I - 2P$. Recall that $P = \hat{v}\hat{v}^T$ from MATH2331, and we now have a formula for H :

$$H = I - 2P = I - 2\hat{v}\hat{v}^T = I - 2\frac{v}{\|v\|}\frac{v^T}{\|v\|} = I - 2\frac{vv^T}{\|v\|^2}$$

Example To find a reflector H to move $x = (2, 2, 1)$ to $w = (3, 0, 0)$, we find v, P, H in order:

$$\begin{aligned} v &= x - w = (2, 2, 1) - (3, 0, 0) = (-1, 2, 1) \\ H &= I - 2\frac{vv^T}{\|v\|^2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{(-1)^2 + 2^2 + 1^2} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 1 & -2 & -1 \\ -2 & 4 & 2 \\ -1 & 2 & 1 \end{bmatrix} \end{aligned}$$

4.2 Important Properties of Householder Reflectors

- H is **symmetric**, $H^T = H$:
 $H = I - 2P$, I is symmetric, and $P = \hat{v}\hat{v}^T$ is symmetric: $(\hat{v}\hat{v}^T)^T = (\hat{v}^T)^T \hat{v} = \hat{v}\hat{v}^T$.
- H is **self-inverse**, $H^{-1} = H$: the inverse of a reflection is itself.
- H is **orthogonal**: $H^{-1} = H = H^T$, implying that $H^T H = I$. Recall from MATH 2331 that a *square* matrix Q is **orthogonal** if and only if $Q^T Q = I$:

$$Q^T Q = \begin{bmatrix} - & q_1^T & - \\ & \vdots & \\ - & q_n^T & - \end{bmatrix} \begin{bmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = I$$

4.3 Householder QR Factorization

We will illustrate the general procedure by an example. For the following matrix, the Householder procedure will produce a QR factorization of the form:

$$\begin{matrix} \begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} & = & \begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{bmatrix} \begin{bmatrix} * & * \\ 0 & * \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \\ A & & Q & R \end{matrix}$$

where Q is an **orthogonal matrix** with orthonormal columns and R is upper-triangular. We break down the algorithm into the following steps.

- 1) Find the first column $w = (*, 0, 0, 0)$ of R so that w has the same length as the first column $x = (1, 1, 1, 1)$ of A . Since $\|x\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2$, we have

$$w = (2, 0, 0, 0)$$
- 2) Find a Householder matrix H_1 to move the first column x of A to the first column w of R by first calculating

$$v = x - w = (1, 1, 1, 1) - (2, 0, 0, 0) = (-1, 1, 1, 1)$$

Then

$$H_1 = I - 2 \frac{vv^T}{\|v\|^2} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

- 3) When we multiply H_1 to A , the first column has the right form as R as expected:

$$H_1 \begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 2 \\ 0 & 2 \\ 0 & 1 \end{bmatrix} \quad vs \quad \begin{bmatrix} * & * \\ 0 & * \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Now we want to find a reflector $\hat{H}_2$ to move $x = (2, 2, 1)$ to $w = (*, 0, 0) = (3, 0, 0)$, as $\|x\| = \sqrt{2^2 + 2^2 + 1^2} = 3$. We did this in Section 4.1 and found

$$\hat{H}_2 = \frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ 2 & -1 & -2 \\ 1 & -2 & 2 \end{bmatrix}$$

- 4) The Householder reflector $H_2 = \begin{bmatrix} 1 & 0 \\ 0 & \hat{H}_2 \end{bmatrix}$ will move $(3, 2, 2, 1)$ to $(3, 3, 0, 0)$, i.e. leaving the first element 3 of the second column unchanged. So, we have:

$$H_2(H_1 A) = H_2 \begin{bmatrix} 2 & 3 \\ 0 & 2 \\ 0 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 3 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = R$$

- 5) Note that $(H_2 H_1)A = R$ implies that

$$A = (H_2 H_1)^{-1} R = H_1^{-1} H_2^{-1} R = H_1 H_2 R = QR$$

where $Q = H_1 H_2$ is an orthogonal matrix as the product of two orthogonal matrices is orthogonal (MATH2331).

$$Q = H_1 H_2 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 2 & 1 \\ 0 & 2 & -1 & -2 \\ 0 & 1 & -2 & 2 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 3 & 5 & -1 & 1 \\ 3 & -1 & 5 & 1 \\ 3 & -1 & -1 & -5 \\ 3 & -3 & -3 & 3 \end{bmatrix}$$

We have finished the QR factorization of A using Householder reflectors:

$$\begin{array}{ccc} \begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} & = \frac{1}{6} \begin{bmatrix} 3 & 5 & -1 & 1 \\ 3 & -1 & 5 & 1 \\ 3 & -1 & -1 & -5 \\ 3 & -3 & -3 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 3 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \\ A & Q & R \end{array}$$

4.4 QR factorization: Householder vs Gram-Schmidt

Before we use the QR factorization to find the least squares solution, there is one more step. Note that R is rectangular $m \times n$ ($m > n$) with an upper triangular $n \times n$ matrix on top and all zero rows below. The zero rows do not contribute to the QR product and we may replace the rectangular R by the top upper triangular $n \times n$ matrix and keep only the first n columns of Q . In our example:

$$\begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 3 & 5 & -1 & 1 \\ 3 & -1 & 5 & 1 \\ 3 & -1 & -1 & -5 \\ 3 & -3 & -3 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 3 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 3 & 5 \\ 3 & -1 \\ 3 & -1 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix}$$

Now we will explain how to find the least squares solution using QR factorization. From the normal equation $(A^T A)x^* = A^T b$ and QR factorization $A = QR$, we have:

$$\begin{aligned} (A^T A)x^* &= A^T b &\Rightarrow ((QR)^T(QR))x^* &= (QR)^T b \\ &&\Rightarrow R^T Q^T Q R x^* &= R^T Q^T b \\ &&\Rightarrow R^T R x^* &= R^T Q^T b \quad (\text{because } Q^T Q = I) \\ &&\Rightarrow (R^T)^{-1} R^T R x^* &= (R^T)^{-1} R^T Q^T b \\ &&\Rightarrow R x^* &= Q^T b \end{aligned}$$

Note that R^T is invertible because it is a square triangular matrix with nonzero diagonal elements, as the vectors in A are independent (homework). **The upshot is that we have replaced the normal equation $(A^T A)x^* = A^T b$ by $Rx^* = Q^T b$.**

Example We will find the least squares solution of the following inconsistent system

$$\begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

using the new method using the QR factorization of A and solving $Rx^* = Q^T b$.

$$\begin{aligned} R &= \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix}, \quad Q = \frac{1}{6} \begin{bmatrix} 3 & 5 \\ 3 & -1 \\ 3 & -1 \\ 3 & -3 \end{bmatrix}, \quad Q^T b = \frac{1}{6} \begin{bmatrix} 3 & 3 & 3 & 3 \\ 5 & -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ Rx^* &= Q^T b \Rightarrow \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \Rightarrow \begin{aligned} x_1 &= 1 \\ x_2 &= 1/3 \end{aligned}$$